Liquidus Phase Diagram of $CaO-SiO_2-La_2O_3-Nb_2O_5$ System with $w(La_2O_3)$ = 15 to 25 Pct

JIYU QIU, CHENGJUN LIU, ZHENGYUE LIU, and DEYING ZHU

The thermodynamic information of the $CaO-SiO_2-Nb_2O_5-La_2O_3$ system is of great significance for the utilization of Nb and RE resources from minerals. In this article, the phase equilibria of the $CaO-SiO_2-Nb_2O_5-La_2O_3$ system were investigated by equilibrium experiments. Twelve kinds of phase equilibria were determined, and then a spatial liquidus phase diagram was constructed. The primary phase fields of SiO_2, $CaNb_2O_6$, $LaNbO_4$, $Ca_2Nb_2O_7$ and $CaSi_3La_4O_{13}$ were determined, respectively. Meanwhile, the liquidus surfaces in the primary phase fields of SiO_2, $Ca_2Nb_2O_7$ and $LaNbO_4$ were determined. Finally, the $CaO-SiO_2-Nb_2O_5$-(17.5, 20, 22.5, 25 pct)La_2O_3 pseudo-ternary phase diagrams were also obtained. In addition to the experimental results, two theoretical rules applicable for the spatial quaternary phase diagram were also proposed to obtain indirect information on the bivariate interface between two adjacent primary phase fields. The first rule was the ''tangent line rule'' for judging related precipitation reaction types, and the second rule was the ''highest temperature rule'' for judging related temperature trends. The present work will be helpful for the utilization of Nb and RE resources and for the establishment of related thermodynamic databases.

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I. INTRODUCTION

In recent years, with the wide application of Nb and RE in the fields of ceramics, steel and other materials,[1–5] the extraction and smelting process of Nb-bearing and RE-bearing minerals has received more attention. The Bayan Obo mineral in China is a typical polymetallic symbiotic mineral, in which there are significant reserves of Nb and RE.[6] However, the complexity of mineral phases and low grade of the mineral composition make the comprehensive utilization of related valuable elements difficult.[4,5] As a result, for many years, most elements except Fe could only be discarded into tailings.[7] Recently, the progress of mineral processing technology has achieved the enrichment of Nb and RE in tailings, which provides a basis for the utilization of these valuable elements.[8,9] However, the development of the related pyrometallurgical process is restricted by the low metal yield and high cost. One important reason is that the coexistence of Nb and RE makes the thermodynamic properties of related pyrometallurgical slag systems complex. According to the composition of related tailings, the $CaO-SiO_2-Nb_2O_5-La_2O_3$ slag system has proved to be a significant fundamental slag system for the improvement of related processes.[10] In terms of the phase diagram, the study of the $CaO-SiO_2-Nb_2O_5-La_2O_3$ system is mainly related to its sub-binary systems and partial sub-ternary systems, including $CaO-SiO_2$,[11] $CaO-La_2O_3$,[12] $Nb_2O_5-La_2O_3$,[13] $CaO-Nb_2O_5$,[14] $SiO_2-La_2O_3$,[15] $SiO_2-Nb_2O_5$,[16] $CaO-SiO_2-La_2O_3$,[17] $CaO-La_2O_3-Nb_2O_5$[18] and $CaO-SiO_2-Nb_2O_5$.[19] To provide practical help for the utilization process of Nb and RE resources and to obtain the original experimental data indispensable for establishing a thermodynamic database, it is necessary to experimentally study the phase equilibria in the $CaO-SiO_2-Nb_2O_5-La_2O_3$ system. For silicate systems with slow phase transformation rates, the equilibrium experiment is widely used to study high-temperature phase equilibria and has proved to be a recognized scientific experimental method.[20–28] In the author's previous work, this method was used for investigating the phase equilibria in the $CaO-SiO_2-Nb_2O_5-La_2O_3$ system with $w(La_2O_3) < 15$ pct, in which the author determined partial liquidus surfaces in the primary phase fields of $CaNb_2O_6$ and SiO_2.[29,30] However, the phase equilibria in the higher $w(La_2O_3)$ region of this quaternary system are still unknown. Therefore, in the current work, the phase equilibria in the $CaO-SiO_2-Nb_2O_5-La_2O_3$ system with $w(La_2O_3)$ = 15 to 25 pct were experimentally studied.

JIYU QIU, CHENGJUN LIU, ZHENGYUE LIU, and DEYING ZHU are with the Key Laboratory for Ecological Metallurgy of Multimetallic Ores (Ministry of Education), and also with the School of Metallurgy, Northeastern University, Shenyang, 110819 Liaoning, China. Contact e-mail: liucj@smm.neu.edu.cn.
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From the perspective of the representation form of phase diagrams, the liquidus phase diagram has a unique advantage in representing the temperature-dependent information. Taking the ternary liquidus projection phase diagram as an example, the univariate line can directly indicate different primary phase fields and indirectly represent the precipitation reaction type. When we use phase diagrams in practice, the above information is helpful for predicting or even controlling the potential primary crystalline phase and solidification structure in materials. However, the rules used in judging the indirect information in the ternary phase diagram, such as the tangent line tule and highest temperature rule,[31] are no longer applicable for the quaternary phase diagram. To obtain the indirect information in the $CaO\text{-}SiO_2\text{-}Nb_2O_5\text{-}La_2O_3$ system, it is necessary to theoretically extend the related rules to a spatial quaternary system.

Table I. Experimental Results of Four-Phase Coexistence

Phase Equilibrium	Typical SEM/XRD Result	Different of Equilibrium Phases in SEM Images
L + S + CN + LN	Fig. 1(a) sample A1 at 1573 K	gray phase → CN white phase → LN black phase → S matrix phase → L
L + CN + C_2N + LN	Fig. 1(b) sample A5 at 1473 K	white bulk phase → LN gray strip phase → CN gray bulk phase → C_2N matrix phase → L
L + CS_3L_2 + C_2N + LN	Fig. 1(c) sample A6 at 1473 K	gray phase → C_2N white regular phase → CS_3L_2 white irregular phase → LN matrix phase → L
L + $C_{10}S_6N$ + C_2N + CS_3L_2	Fig. 1(d) sample A8 at 1473 K	black phase → $C_{10}S_6N$ gray strip phase → C_2N white phase → CS_3L_2 residual matrix phase → L
L + LN + S + CS_3L_2	Fig. 1(e) sample A4 at 1473 K	white strip phase → LN white hexagonal phase → CS_3L_2 black phase → S matrix phase → L

Table II. Experimental Results of Three-Phase Coexistence

Phase Equilibrium	Typical SEM/XRD Result	Different of Equilibrium Phases In SEM Images
L + C_2N + LN	Fig. 2(a) sample B3 at 1573 K	white phase → LN gray phase → C_2N matrix phase → L
L + C_2N + CS_3L_2	Fig. 2(b) sample B5 at 1473 K	gray phase → C_2N white phase → CS_3L_2 matrix phase → L
L + LN + S	Fig. 2(c) + Fig. 4(a) sample B2 at 1573 K	black phase → S white phase → LN matrix phase → L
L + CN + S	Fig. 2(d) sample B1 at 1573 K	black phase → S gray strip phase → CN matrix phase → L.

Table III. Experimental Results of Two-Phase Coexistence

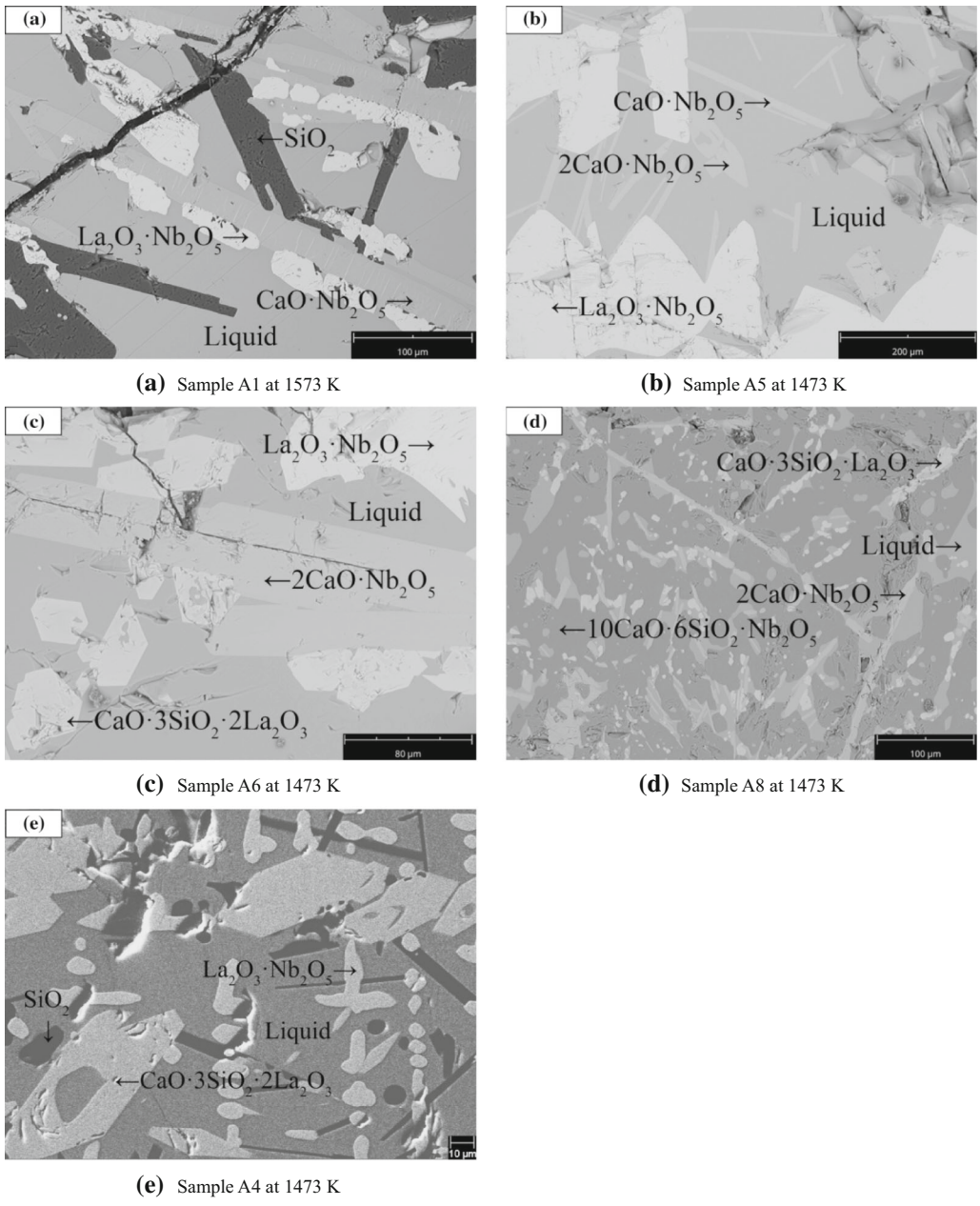
Phase Equilibrium	Typical SEM/XRD Result	Different of Equilibrium Phases in SEM Images
L + LN	Fig. 3(a) + Figure 4(b) sample C8 at 1573 K	white phase → LN matrix phase → L
L + C_2N	Fig. 3(b) sample C6 at 1573 K	white phase → C_2N matrix phase → L.
L + S	Fig. 3(c) sample C1 at 1473 K	black phase → S matrix phase → L

Table IV. Phase Equilibria of CaO-SiO_2-Nb_2O_5-La_2O_3 -System and Related Composition of Equilibrium Phase, Mass Pct

Slag No.	No. in Phase Diagram	T, K	Phase Equilibria	Phase	CaO	SiO_2	Nb_2O_5	La_2O_3
A1	3	1573	L + S + CN + LN	L	12.83 pct	27.31 pct	39.56 pct	20.30 pct
				S	—	100 pct	—	—
				CN	17.55 pct	—	82.45 pct	—
				LN	—	—	44.76 pct	45.24 pct
A2	4	1473	L + S + CN + LN	L	19.74 pct	35.30 pct	26.40 pct	18.56 pct
				CN	17.42 pct	—	82.58 pct	—
				S	—	100.00 pct	—	—
				LN	—	—	44.72 pct	55.28 pct
A3	11	1573	L + C_2N + LN + CS_3L_2	L	18.06 pct	16.13 pct	39.73 pct	26.08 pct
				C_2N	18.76 pct	—	70.66 pct	10.57 pct
				$CS3L_2$	6.01 pct	16.62 pct	7.88 pct	69.49 pct
				LN	—	—	44.68 pct	55.32 pct
A4	2	1473	L + CS_3L_2 + LN + S	L	16.83 pct	33.54 pct	25.33 pct	24.30 pct
				CS_3L_2	7.07 pct	19.69 pct	—	73.24 pct
				LN	—	—	44.79 pct	55.21 pct
				S	—	100.00 pct	—	—
A5	9	1473	L + CN + C_2N + LN	L	25.16 pct	28.98 pct	28.65 pct	17.21 pct
				CN	17.39 pct	—	82.61 pct	—
				C_2N	21.67 pct	—	70.44 pct	7.89 pct
				LN	—	—	45.01 pct	54.99 pct
A6	10	1473	L + C_2N + LN + CS_3L_2	L	20.22 pct	21.28 pct	33.08 pct	25.42 pct
				C_2N	21.32 pct	—	70.86 pct	7.82 pct
				CS_3L_2	8.18 pct	17.67 pct	10.97 pct	63.19 pct
				LN	1.94 pct	2.16 pct	42.68 pct	53.22 pct
A7	10	1473	L + C_2N + LN + CS_3L_2	L	20.08 pct	20.97 pct	33.14 pct	25.81 pct
				C_2N	21.73 pct	—	70.52 pct	7.75 pct
				CS_3L_2	8.68 pct	17.69 pct	10.59 pct	63.04 pct
				LN	1.92 pct	2.04 pct	42.58 pct	53.45 pct
A8	21	1473	L + CS_3L_2 + $C_{10}S_6N$ + C_2N	L	23.64 pct	25.62 pct	30.54 pct	20.21 pct
				CS_3L_2	8.30 pct	19.27 pct	6.36 pct	66.07 pct
				$C_{10}S_6N$	39.38 pct	29.56 pct	31.06 pct	—
				C_2N	21.51 pct	—	70.99 pct	7.50 pct
B1	34	1573	L + S + CN	L	16.11 pct	28.62 pct	38.49 pct	16.78 pct
				S	—	100 pct	—	—
				CN	17.50 pct	—	82.50 pct	—
B2	1	1573	L + LN + S	L	7.55 pct	25.50 pct	36.83 pct	30.13 pct
				LN	—	—	44.67 pct	55.33 pct
				S	—	100 pct	—	—
B3	12	1573	L + C_2N + LN	L	18.66 pct	16.91 pct	39.78 pct	24.64 pct
				C_2N	18.81 pct	—	70.05 pct	11.13 pct
				LN	—	—	45.03 pct	54.97 pct
B4	17	1573	L + C_2N + CS_3L_2	L	19.46 pct	18.01 pct	37.42 pct	25.11 pct
				C_2N	19.42 pct	—	70.42 pct	10.16 pct
				CS_3L_2	6.42 pct	16.55 pct	7.86 pct	69.16 pct
B5	18	1473	L + C_2N + CS_3L_2	L	21.89 pct	23.37 pct	31.82 pct	22.91 pct
				C_2N	21.38 pct	—	70.89 pct	7.73 pct
				CS_3L_2	8.31 pct	17.94 pct	9.56 pct	64.18 pct
B6	20	1473	L + C_2N + CS_3L_2	L	22.59 pct	24.27 pct	31.31 pct	21.83 pct
				C_2N	21.60 pct	—	70.97 pct	7.43 pct
				CS_3L_2	8.16 pct	18.27 pct	8.07 pct	65.49 pct
C1	48	1573	L + S	L	22.35 pct	35.57 pct	30.97 pct	11.11 pct
				S	—	100.00 pct	—	—
C2	49	1473	L + S	L	21.64 pct	37.09 pct	24.68 pct	16.58 pct
				S		100.00 pct		
C3	50	1473	L + S	L	29.72 pct	44.75 pct	17.74 pct	7.79 pct
				S	—	100.00 pct	—	—
C4	51	1473	L + S	L	32.31 pct	50.98 pct	10.11 pct	6.60 pct
				S	—	100.00 pct	—	—
C5	52	1473	L + S	L	35.82 pct	56.02 pct	5.44 pct	2.72 pct
				S	—	100.00 pct	—	—
C6	26	1573	L + C_2N	L	27.76 pct	22.76 pct	38.38 pct	11.10 pct
				C_2N	22.52 pct	—	71.05 pct	6.43 pct

Table IV. continued

Slag No.	No. in Phase Diagram	T, K	Phase Equilibria	Phase	CaO	SiO_2	Nb_2O_5	La_2O_3
C7	27	1573	L + C_2N	L	25.79 pct	20.91 pct	37.79 pct	15.52 pct
				C_2N	20.40 pct	—	72.61 pct	6.99 pct
C8	13	1573	L + LN	L	17.46 pct	16.48 pct	39.48 pct	26.58 pct
				LN	1.75 pct	2.16 pct	42.44 pct	53.65 pct



(a) Sample A1 at 1573 K

(b) Sample A5 at 1473 K

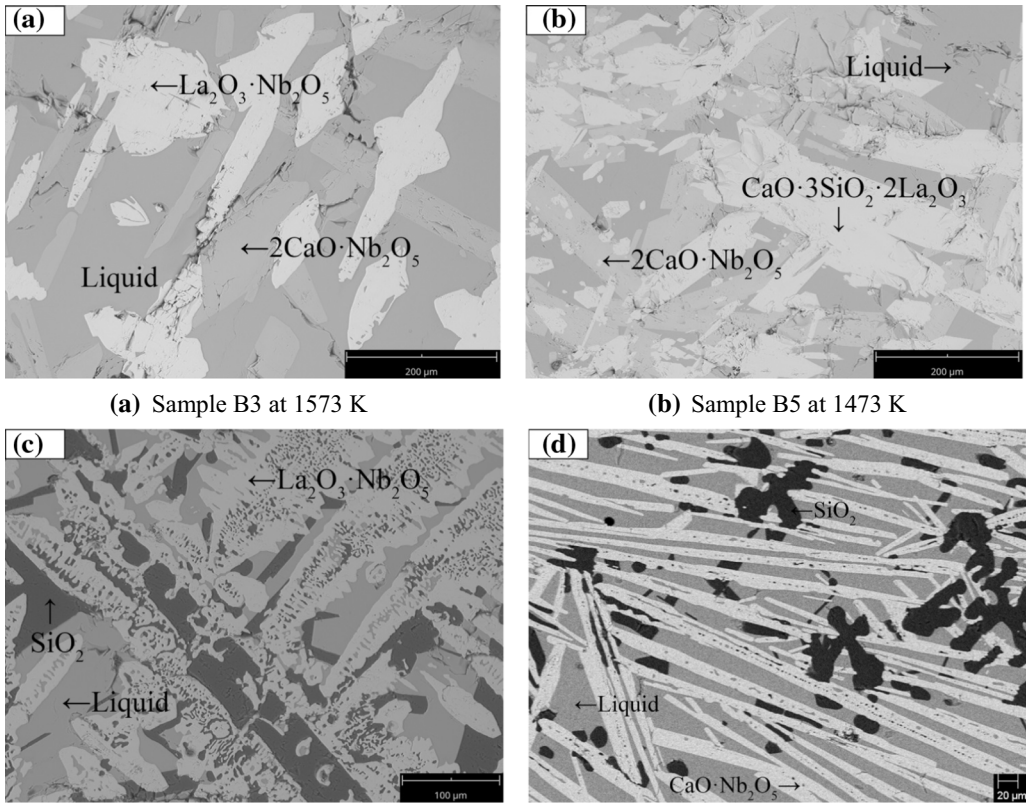
(c) Sample A6 at 1473 K

(d) Sample A8 at 1473 K

(e) Sample A4 at 1473 K

Fig. 1—Typical SEM results of the samples with four-phase coexistence.

In the current work, equilibrium experiments were performed to study the phase equilibria in the CaO-SiO_2-Nb_2O_5-La_2O_3 system. Scanning electron microscope (SEM), energy-dispersive spectrometer (EDS) and X-ray diffraction (XRD) analyses were used to identify the phase equilibria and to measure the composition of each equilibrium phase. According to the experimental results in the current and previous work, the spatial



(a) Sample B3 at 1573 K **(b)** Sample B5 at 1473 K

(c) Sample B2 at 1573 K **(d)** Sample B1 at 1573 K

Fig. 2—Typical SEM results of the samples with three-phase coexistence.

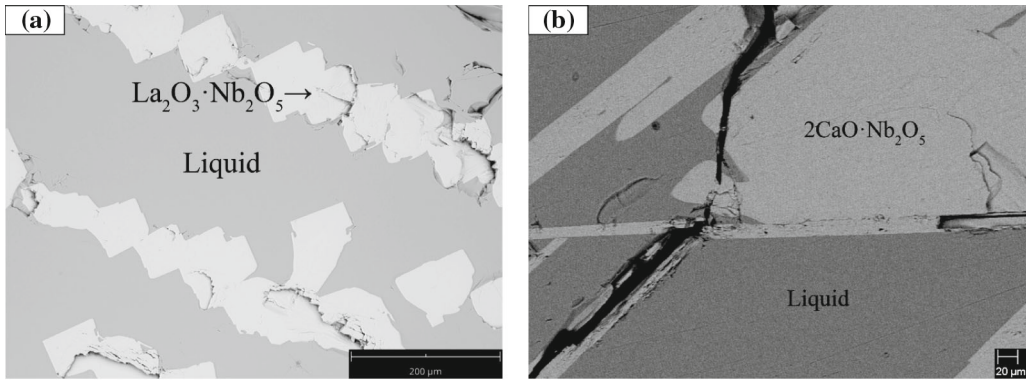
quaternary system was constructed. Then, the CaO-SiO_2-Nb_2O_5-(17.5, 20, 22.5, 25 pct)La_2O_3 pseudo-ternary systems were obtained for convenient use, respectively. In addition, considering the tangent line and highest temperature rules are no longer suitable for the spatial quaternary system, corresponding applicable rules were deduced to judge the precipitation reaction types and temperature trends on the bivariate interface in the quaternary system. The experimental results in this article will be helpful for the utilization of Nb and RE resources and for further establishment of relevant thermodynamic databases.

A. *Experimental Procedure*

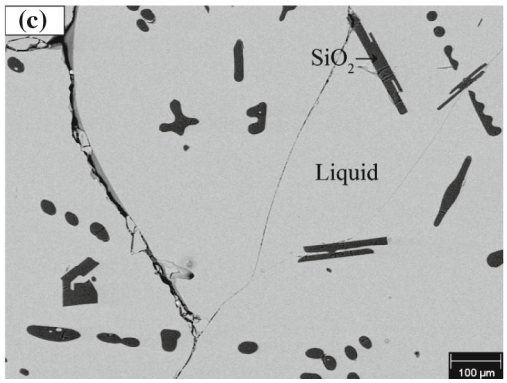
To get reliable experimental results, the samples used in the equilibrium experiment were pre-melted to achieve a homogeneous composition. Details of the pre-melt process were described in a previous work.[32] One gram pre-melted sample was placed in a platinum crucible in a $MoSi_2$ furnace and then heated to 1873 K and kept for 10,800 seconds (3 hours). Subsequently, the sample was cooled to experimental temperature and kept for 86,400 seconds (24 hours); for some samples, the equilibrium time was 172,800 seconds (48 hours). After that, the sample was quenched with ice water to maintain the phase equilibrium at high temperature. SEM (EVO18 type produced by Zess), EDS (X-Max20 type produced by Oxford) and XRD (X' pert PRO type produced by PANalytical) were then used to identify the phase equilibrium in each sample and to measure the composition of each equilibrium phase.

II. RESULTS

In the current work, 22 samples were used in the equilibrium experiment. According to the SEM-EDS and XRD results, 12 kinds of phase equilibria were determined, including 5 kinds of four-phase coexistence, 4 kinds of three-phase coexistence and 3 kinds of two-phase coexistence. Tables I through III show the SEM/XRD results of four-, three- and two-phase coexistence, respectively. Table IV shows all the current phase equilibria of the samples and the related composition of each equilibrium phase. Figures 1 through 3 show the SEM result of typical samples of four-, three- and two-phase coexistence, respectively Figure 4.

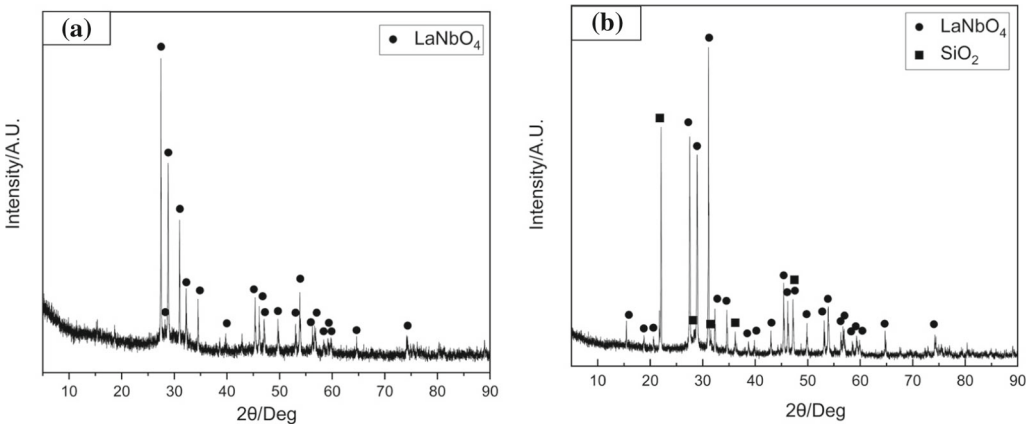


(a) Sample C8 at 1573 K **(b)** Sample C6 at 1573 K



(c) Sample C1 at 1473 K

Fig. 3—Typical SEM results of the samples with two-phase coexistence.



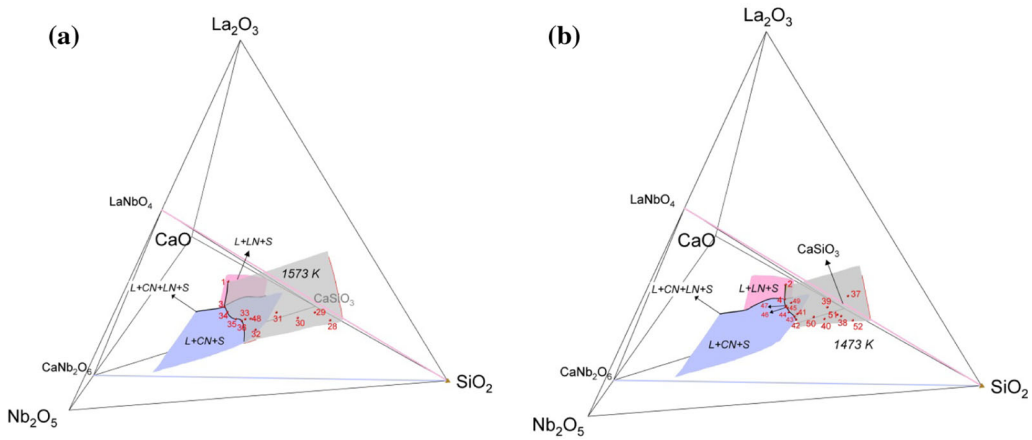
(a) Sample B2 at 1573 K **(b)** Sample C8 at 1573 K

Fig. 4—Typical XRD results of samples.

III. DISCUSSION

A. *Spatial Phase Diagram of the CaO-SiO_2-Nb_2O_5-La_2O_3 System with $w(La_2O_3)$ = 15 to 25 pct*

Based on the Gibbs phase rule, the general spatial representation form of a quaternary phase diagram was confirmed in the previous work.[29,30] According to the phase equilibria in Table IV, a spatial phase diagram of the CaO-SiO_2-Nb_2O_5-La_2O_3 quaternary system was constructed. It should be pointed out that, due to the continuity of the phase diagram, the phase equilibria in the CaO-SiO_2-Nb_2O_5-La_2O_3 system with $w(La_2O_3) <$ 15 pct and the phase equilibria in the CaO-SiO_2-La_2O_3 system should also be considered in the construction process. Figures 5 through 7 show the constructed CaO-SiO_2-Nb_2O_5-La_2O_3 phase diagram with $w(La_2O_3)$ = 15 to 25 pct. The details of the numbered liquid phase in Figures 5 through 7 are given in Table IV (current work) and Table V (previous work).



(a) Part 1 **(b)** Part 2

Fig. 5—Spatial phase diagram of the CaO-SiO_2-Nb_2O_5-La_2O_3 system in the low basicity region within the temperature range of 1473 to 1573 K, mass pct.

Table V. Phase Equilibria of CaO-SiO_2-Nb_2O_5-La_2O_3-System from Reference

No. in Phase Diagram	Sample No. in Reference	T, K	References	Phase Equilibria	No. in Phase Diagram	Sample No. in Reference	T, K	References	Phase Equilibria
5	A7	1473	29	L + CN + LN	33	A8	1573	30	L + S
6	A11	1573	29	L + CN + LN	35	A11	1573	30	L + CN + S
7	A10	1573	29	L + CN + LN	36	B6	1573	30	L + CN + S
8	B7	1573	29	L + CN + C_2N + LN	37	A1	1473	30	L + S
16	A6	1573	29	L + CN + LN	38	A3	1473	30	L + S
19	A5	1473	29	L + CN + LN	39	A4	1473	30	L + S
22	B1	1473	29	L + CN + C_2N	40	A5	1473	30	L + S
23	B5	1473	29	L + CN + C_2N	41	A6	1473	30	L + S
24	B3	1573	29	L + C_2N	42	B2	1473	30	L + CN + S
25	B2	1573	29	L + C_2N	43	A10	1473	30	L + CN + S
28	A2	1573	30	L + C_2N	44	B5	1473	30	L + CN + S
29	A3	1573	30	L + C_2N	45	A11	1473	30	L + CN + S
30	A5	1573	30	L + S	46	B6	1473	30	L + CN + S
31	A6	1573	30	L + S	47	B1	1473	30	L + CN + S
32	A7	1573	30	L + S					

"w(CaO)/w(SiO_2), also known as basicity," is a widely used parameter in practical application. The phase diagram will be described for different basicity regions.

Figure 5(a) shows the lower basicity region of the CaO-SiO_2-Nb_2O_5-La_2O_3 phase diagram, in which the purple surface is the bivariate interface between the primary phase fields of CN and S, the pink surface is the bivariate interface between the primary phase fields of LN and S, and the gray surface is the liquidus surface at 1573 K in the primary phase field of S. Figure 5(b) shows the liquidus surface at 147 K in the primary phase field of S. They were constructed based on the previous work[29,30] and present new data. As can be seen, the bivariate interface of LN-S is almost perpendicular to the CaO-SiO_2-Nb_2O_5 system, which means that with the increase of w(La_2O_3) the potential primary phase field of LN will shrink and the primary phase field of S will expand.

Figure 6 shows the medium basicity region of the CaO-SiO_2-Nb_2O_5-La_2O_3 phase diagram, in which the purple surface is the bivariate interface between the primary phase fields of C_2N and CN, the cyan surface is the bivariate interface between the primary phase fields of C_2N and LN and the orange surface the bivariate interface between the primary phase fields of C_2N and CS_3L_2. Additionally, the two green surfaces are liquidus surfaces at 1473 and 1573 K in the primary phase field of LN. The phase diagram in the upper left corner of Figure 6 shows that the primary phase field of LN has a slender strip shape rather than the wide shape of the primary phase field of S.

Figure 7 shows the higher basicity region of the CaO-SiO_2-Nb_2O_5-La_2O_3 phase diagram, in which the orange surface is the bivariate interface between the primary phase fields of CS_3L_2 and C_2N, and the green surfaces are the liquidus surfaces at 1473 and 1573 K in

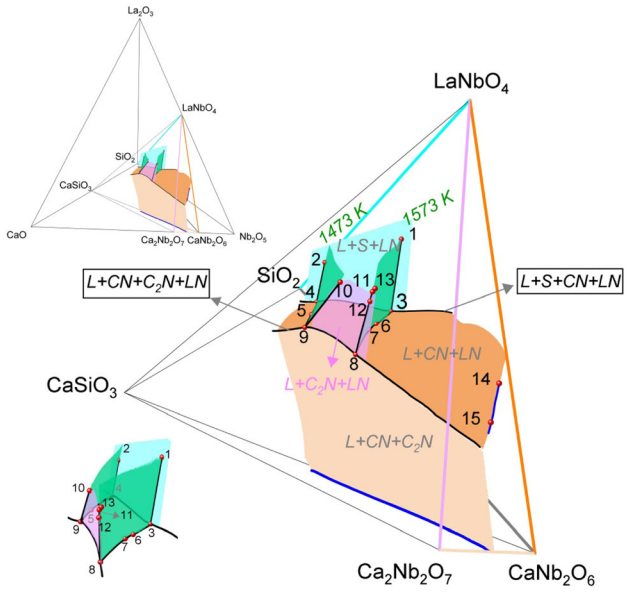


Fig. 6—Spatial phase diagram of the CaO-SiO_2-Nb_2O_5-La_2O_3 system in the medium basicity region within the temperature range of 1473 to 1773 K, mass pct.

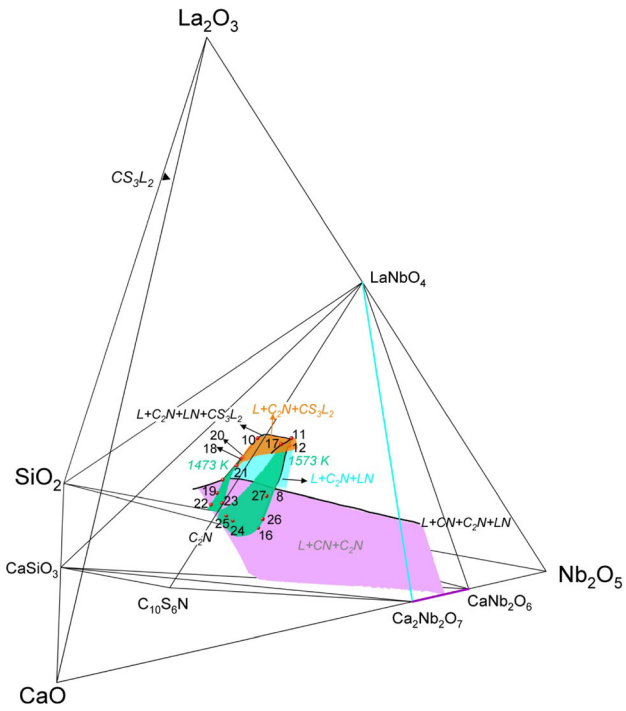


Fig. 7—Spatial phase diagram of the high basicity region in the CaO-SiO_2-Nb_2O_5-La_2O_3 system, mass pct.

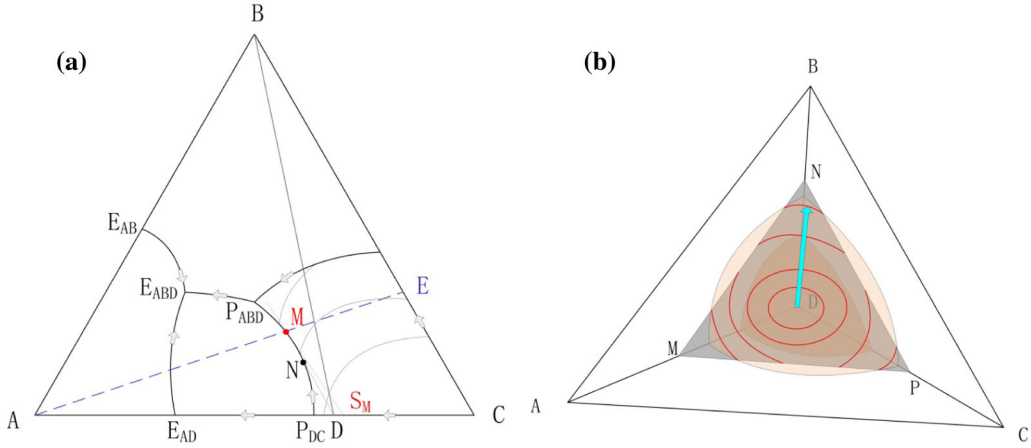
the primary phase field of C_2N. The liquidus surfaces are constructed based on the previous work and present new data. As shown in Figure 7, the $w(La_2O_3)$ upper limit of the primary phase field of C_2N is about 25 pct, *i.e.*, with more $w(La_2O_3)$, the primary phase field of C_2N will disappear and the primary phase field of CS_3L_2 will appear. It can be inferred that the potential bivariate interface of LN-CS_3L_2 will appear when there is more $w(La_2O_3)$ in the starting composition. Meanwhile, the bivariate interface of LN-CS_3L_2 will share the univariate line of L + TC_2N + LN + CS_3L_2 with the bivariate interface of C_2N-LN and the bivariate interface of C_2N-CS_3L_2. Phase equilibria in the higher $w(La_2O_3)$ region of CaO-SiO_2-Nb_2O_5-La_2O_3 phase diagram will be investigated in the future.

B. *Tangent Line Rule and Highest Temperature Rule Applicable for the Quaternary Phase Diagram*

As described above, the traditional tangent line and highest temperature rules are no longer applicable for the bivariate interface corresponding to three-phase coexistence in the quaternary phase diagram. Therefore, to evaluate the precipitation reaction type and temperature variation on the bivariate interface in the CaO-SiO_2-Nb_2O_5-La_2O_3 system, the two rules were modified and improved to be applicable for the quaternary phase diagram.

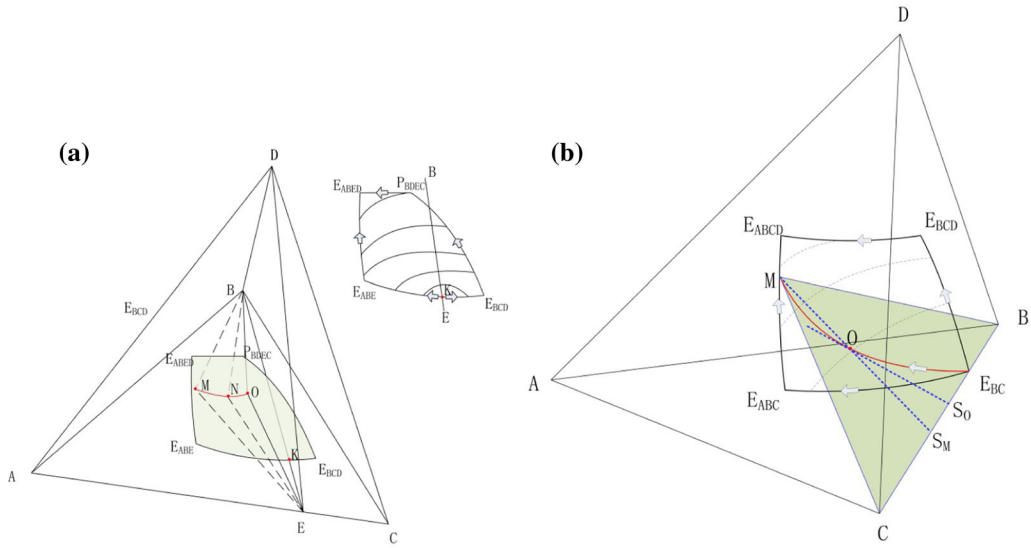
Before making deductions, the application of the two rules in the ternary phase diagram is briefly described. As Figure 8(a) shows, for any tangent line of the univariate line between two adjacent primary phase fields, there is an intersection point on the corresponding sub-solidus line segment. According to the tangent line rule, the precipitation reaction type can be evaluated based on the relative position of the intersection point and the sub-solidus line segment, such as the eutectic curve E_{AB}-E_{ABD}, peritectic curve N-P_{DC} and eutectic curve N-P_{BCD}. In addition, the temperature trend on the univariate line can be judged by the the highest temperature rule, such as $P_{ABD} \rightarrow E_{ABD}$ being the direction of single temperature decline. It should be pointed out that the two rules mainly depend on the real relative position of relevant thermodynamic information, which means it is impossible to apply them in a pseudo-phase diagram. For example, the precipitation reaction type of point M on curve P_{ABD}-P_{CD} cannot be judged only in the A-E pseudo-binary system. Besides, even a liquidus temperature distribution with the characteristic of higher central temperature may occur in some pseudo-binary systems (such as the M-E pseudo-binary system). For the quaternary system, even the ''onion slice'' with the characteristic of higher central liquidus temperature may occur in some pseudo-ternary sections, as shown in Figure 8(b). Therefore, to promote the application of the quaternary phase diagram, the tangent line and highest temperature rules, which are suitable for the quaternary system, were deduced in the present work, as follows.

Highest temperature rule: The conclusion is that the intersection point of the sub-solidus line segment and the corresponding bivariate interface has the highest liquidus temperature on the bivariate interface. Assuming that there is a sample with starting composition at point O on the bivariate interface of B-E, when it shows three-phase coexistence at temperatures T_M and T_N, respectively, the composition of the corresponding liquid phase L_M and L_N will be located at the intersection curve of plane B-O-E and the bivariate interface of



(a) A-B-C ternary system **(b)** Pseudo-ternary phase diagram with "onion slice"

Fig. 8—Schematic phase diagrams for the tangent line rule and highest temperature rule.



(a) Schematic diagram for "Highest Temperature Rule" **(b)** Schematic diagram for "Tangent Line Rule"

Fig. 9—Tangent line rule and highest temperature rule in the quaternary system.

B-E, namely point M and N in Figure 9(a). According to the spatial lever rule, the proportion of liquid phase at T_M is the area ratio of ΔBOE and ΔBME and that at T_N is the area ratio of ΔBOE and ΔBNE. As $S_{\Delta BNE}$ is larger than $S_{\Delta BME}$, the proportion liquid phase at point M will be lower than that at point N. For any point on the bivariate interface of B-E, the farther it is from K point, the lower liquidus temperature it will have. That is the highest temperature rule suitable for the quaternary phase diagram.

Tangent line rule: As shown in Figure 9(b), curve M-O-E_{BC} is the intersection curve of plane B-O-C and the bivariate interface of B-C, the straight line OS_O is its tangent line through point O, and the straight line OS_M represents one of its secant lines through point O. Assuming that there is a sample with starting composition at point O, when the temperature is T_M, the equilibrium liquid composition will be located at point M, and the total solid composition point will be located at point S_M. When discussing equilibrium temperature changes from T_M to T_O along the univariate curve M-O-E_{BC} on the bivariate interface of B-C, the total solid composition will change from S_M to S_O along the straight line segment BC. This means that the S_O is the instantaneous total precipitation composition at point O. According to the relative position of point S_O and segment line BC, the precipitation reaction type can be judged as a eutectic or peritectic reaction. For a quaternary system, the intersection point S_O is also on the tangent plane of the bivariate interface of B-C through point O, which means the tangent line rule can also be called the tangent plane rule.

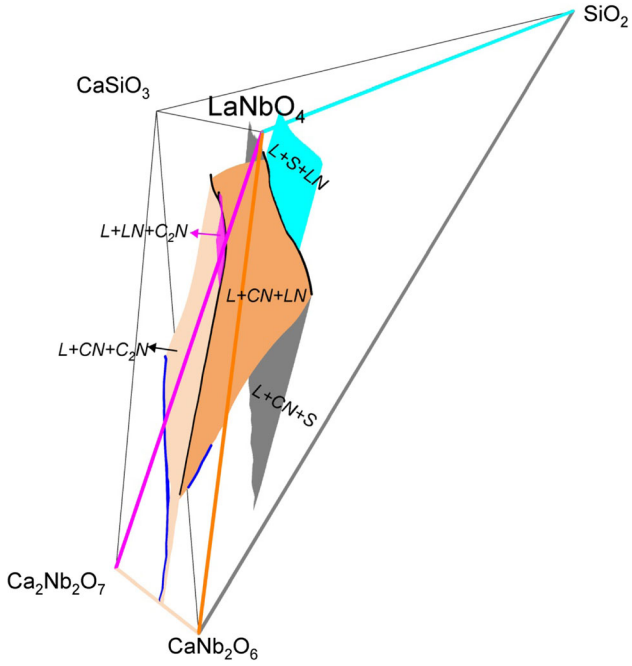
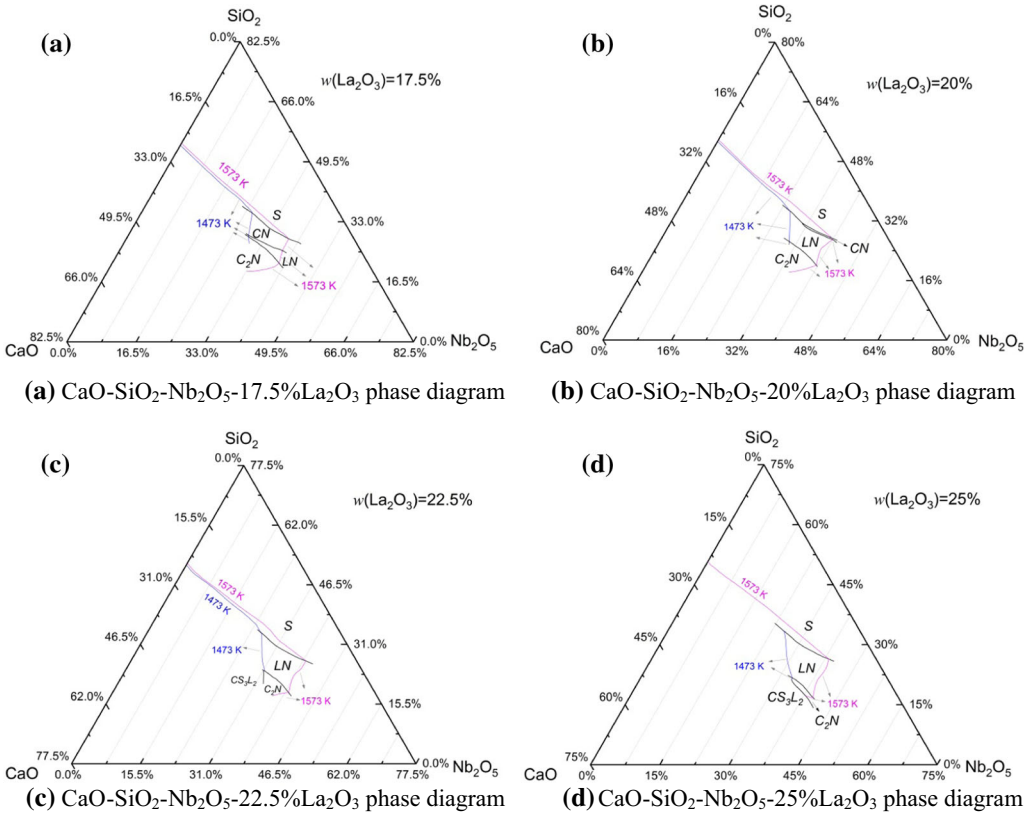


Fig. 10—Precipitation reaction type of the bivariate interface in the $CaO\text{-}SiO_2\text{-}Nb_2O_5\text{-}La_2O_3$ system, mass pct [basicity = 0.5 to 2, $w(Nb_2O_5)$ = 20 to 80 pct, $w(La_2O_3) < 30$ pct); experimental temperature range from 1473 to 1773 K].

Based on the tangent line rule, the precipitation reaction type of some bivariate interfaces in the $CaO\text{-}SiO_2\text{-}Nb_2O_5\text{-}La_2O_3$ phase diagram can be judged. As shown in Figure 10, under the experimental composition and temperature range, the bivariate interfaces of CN-LN, LN-S, C_2N-LN and CN-S are all judged to be eutectic reactions with type $L \rightarrow Solid_1 + Solid_2$. Similarly, the bivariate interface of $C_2N\text{-}CS_3L_2$ can be considered a eutectic interface. Considering the short line segment of $CN\text{-}C_2N$ and solid solution of C_2N, its precipitation reaction type should be cautiously determined after supplying more experimental data in the future.

C. *$CaO\text{-}SiO_2\text{-}Nb_2O_5$-(17.5, 20, 22.5, 25 pct)La_2O_3 Pseudo-ternary Phase Diagram*

Although the spatial phase diagram can visually represent the real phase equilibria in a quaternary system, it is not convenient for direct use. Therefore, based on the constructed spatial phase diagrams shown in Figures 5 through 7, the $CaO\text{-}SiO_2\text{-}Nb_2O_5$ pseudo-ternary phase diagrams with $w(La_2O_3)$ = 17.5, 20, 22.5 and 25 pct were also obtained, respectively, as shown in Figures 11(a) through (d). The involved primary phase fields and liquidus lines are consistent with the spatial phase diagram, including S, CN, LN, C_2N and CS_3L_2. As can be seen, in the region where $w(La_2O_3)$ is < 15 pct, with the increase of basicity, the primary phase field



(a) $CaO\text{-}SiO_2\text{-}Nb_2O_5$-17.5%$La_2O_3$ phase diagram **(b)** $CaO\text{-}SiO_2\text{-}Nb_2O_5$-20%$La_2O_3$ phase diagram

(c) $CaO\text{-}SiO_2\text{-}Nb_2O_5$-22.5%$La_2O_3$ phase diagram **(d)** $CaO\text{-}SiO_2\text{-}Nb_2O_5$-25%$La_2O_3$ phase diagram

Fig. 11—Pseudo-ternary phase diagrams of the $CaO\text{-}SiO_2\text{-}Nb_2O_5$ system with different $w(La_2O_3)$, mass pct (experimental temperature from 1473 to 1573 K) *Abbreviation* CaO→ C, $SiO_2 \rightarrow$ S, $Nb_2O_5 \rightarrow$ N, $La_2O_3 \rightarrow$ L.

will follow the change of S→ CN→ C_2N. In the region where $w(La_2O_3)$ is 15 to 20 pct, with the increase of basicity, the change of the primary phase field is S→ CN→ LN→ C_2N. In the region where $w(La_2O_3)$ is about 25 pct, the change of the primary phase field is S→ LN→ C_2N→ CS_3L_2.

With the exception of the liquidus surface information in the primary phase region of CN, which was taken from the literature,[29] all other information was determined by combining the previous work and the current experimental data. More experiments will be carried out in further experiments to determine the phase equilibria in the higher $w(La_2O_3)$ region of the CaO-SiO_2-Nb_2O_5-La_2O_3 system.

IV. CONCLUSIONS

In the current work, the phase equilibria in the CaO-SiO_2-Nb_2O_5-La_2O_3 system with $w(La_2O_3)$ = 15 to 25 pct were studied. In terms of experiments, the liquidus phase diagram was constructed, including the primary phase fields of S, CN, LN, C_2N and CS_3L_2. The liquidus surfaces at 1473 and 1573 K in the primary phase fields of S, LN and C_2N were determined. Finally, the CaO-SiO_2-Nb_2O_5 psudo-ternary system phase diagrams with $w(La_2O_3)$ = 17.5, 20, 22.5 and 25 pct were obtained for convenient use. In terms of theory, the tangent line rule and highest temperature rule, which are applicable for the quaternary system, were deduced to judge the precipitation reaction type and temperature trend on the bivariate interface. We hope the present work will be helpful for the utilization of Nb and RE resources and development of quaternary system phase diagrams.

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